Progressive Visual Analytics of Incident Logs: Streaming and Forensic Analysis

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***Abstract*— Progressive visual analytics enables the interactive exploration of large and streaming datasets by providing incremental insights without requiring full computation. In this work, we use the ProgressiVis framework to analyze the Incident Management Process Enriched Event Log dataset using both streaming and forensic analytical approaches. For the streaming analysis, we examine the `made_sla` attribute to identify SLA breaches, visually investigate contributing factors, and propose actionable measures to reduce future violations. In addition, we perform temporal trend analysis to identify peak incident periods across hours and days, enabling more effective workforce allocation and resource planning.For the forensic analysis, we conduct a retrospective structural examination of historical incident records to analyze escalation depth, reassignment patterns, resolution times, and the relationship between SLA compliance, incident priority, and incident complexity. This analysis facilitates the identification of structural inefficiencies, behavioral patterns, and operational anomalies within incident resolution workflows. Collectively, these approaches demonstrate how progressive visual analytics can provide timely operational insights from streaming data while also enabling deeper structural diagnosis to support long-term operational optimization and informed decision-making.**

I. INTRODUCTION

The efficient management of IT incidents is crucial for maintaining service quality and meeting organizational service-level agreements (SLAs). The **Incident Management Process Enriched Event Log** provides a rich dataset for analyzing incident resolution patterns and understanding factors that influence SLA compliance. In this work, we employ progressive visual analytics to study both streaming and forensic aspects of incident data, aiming to uncover actionable insights that can improve operational performance.

For the streaming analysis, we first focus on the attribute $made_sla$, a boolean field indicating whether an incident met its target SLA. Our approach starts by assessing whether a higher number of distinct incidents exceeds the target SLA or remains within it. We then use heatmaps to visually explore how incident attributes such as $priority$, $impact$, and $urgency$ affect SLA compliance, identifying which types of incidents require immediate attention. Subsequently, we investigate whether the assigned support group impacts SLA breaches, analyzing potential bottlenecks by defining a capability score to quantify group effectiveness. Further, we examine the top five incident $categories$ and their $priority$ distributions, providing insight into areas that require focused intervention. Finally, we assess the influence of $knowledge_base$ usage on SLA outcomes, comparing the odds of meeting SLA targets when knowledge base resources are utilized versus when they are not.

In addition to SLA analysis, trend analysis is performed to detect temporal patterns in incident occurrences. We categorize incidents by day_of_week to identify peak periods and examine hourly patterns to detect times of high incident load. Differences between weekdays and weekends are analyzed to generalize patterns across time. Moreover, we explore category-specific trends across different time slots and investigate whether incident volume impacts SLA breaches, helping to identify potential load-induced bottlenecks.

In addition to progressive streaming analysis, we perform a forensic analysis to uncover structural patterns and operational inefficiencies. Unlike streaming analysis, which focuses on incremental insight generation, forensic analysis enables retrospective examination of incident resolution workflows. Specifically, we analyze resolution time computed from the difference between `opened_at` and `closed_at` timestamps, and examine how escalation-related factors such as reassignment count, reopen count, and modification frequency influence resolution efficiency and SLA compliance. We also investigate the relationship between incident severity attributes (priority, impact, and urgency) and resolution outcomes, to determine whether higher-severity incidents receive faster and more effective handling as expected. Additionally, clustering and anomaly detection techniques are used to identify latent behavioral patterns and anomalous operational periods. This analysis provides deeper structural insights into incident management performance and helps identify opportunities for improving operational efficiency, escalation handling, and SLA adherence.

Here, we adopt a free-style coding approach, leveraging

snippets of ProgressiVis to create powerful custom widgets and charts combined with Plotly. Since existing modules offer limited customizability, this approach allows us to tailor visualizations precisely to our analysis needs.

Together, these analyses demonstrate how progressive visual analytics can provide both operational and strategic insights from dynamic incident datasets, supporting informed decision-making and proactive resource management.

II. RELATED WORK

Prior research has applied analytical and machine learning techniques to improve incident management efficiency. In [1], Lim and Song use the K-Means clustering algorithm on incident management logs to group incidents based on attributes such as urgency, impact, and reassignment count. Their analysis identifies distinct operational patterns, including high-volume low-priority incidents and smaller clusters of complex incidents, enabling more effective prioritization and improved incident response time.

Similarly, in [2], Anggraeni et al. propose a graph-based analytical framework to model relationships between support groups using incident transfer data. By constructing a work relationship network and applying Graph Neural Networks, the study identifies substitute human resources capable of handling similar incident types, improving workforce flexibility and operational continuity.

Unlike these works, which primarily focus on clustering, prediction, or structural modeling, our approach emphasizes progressive visual analytics using the ProgressiVis framework to incrementally derive actionable insights from streaming incident data. This enables interactive exploration of SLA compliance, temporal patterns, and operational inefficiencies, supporting real-time decision-making without requiring complete offline computation.

III. ANALYSIS OF STREAMING INCIDENT DATA

A. SLA Compliance Analysis

1) Overall SLA Compliance: To begin our investigation of SLA compliance, we focus on the attribute $made_sla$, which is a boolean attribute that shows whether the incident exceeded the target SLA. Each incident in the dataset may contain multiple event records, and the target SLA can evolve during the incident lifecycle. Since our primary concern is the final outcome, we aggregate records at the incident level and classify each incident based on whether the SLA was ultimately met ($made_sla = False$) or breached ($made_sla = True$).

To examine this visually, we construct a bar chart where the x-axis represents the two possible outcomes—SLA met and SLA breached, and the y-axis represents the number of unique incidents in each category. This visualization provides a clear overview of the overall SLA compliance distribution and allows us to immediately identify any imbalance between successful resolutions and SLA violations.

To reflect the progressive nature of the analysis and communicate reliability, we visualize uncertainty in SLA counts as data is processed incrementally. The chart is refreshed in batches of 50 incidents.

Let $C_{current,met}$ and $C_{previous,met}$ denote the cumulative number of unique incidents that met the SLA in the current and previous batches, respectively. Similarly, $C_{current,unmet}$ and $C_{previous,unmet}$ represent the cumulative counts of SLA-breached incidents.

The uncertainties are computed as:

$$uncertainty_{met} = |C_{current,met} - C_{previous,met}|$$
$$uncertainty_{unmet} = |C_{current,unmet} - C_{previous,unmet}|$$

These values are displayed as error bars. When the uncertainties remain consistently low (e.g., 0–5) across batches, it indicates convergence and allows reliable conclusions to be drawn, enabling informed operational decisions even before the full dataset has been processed.

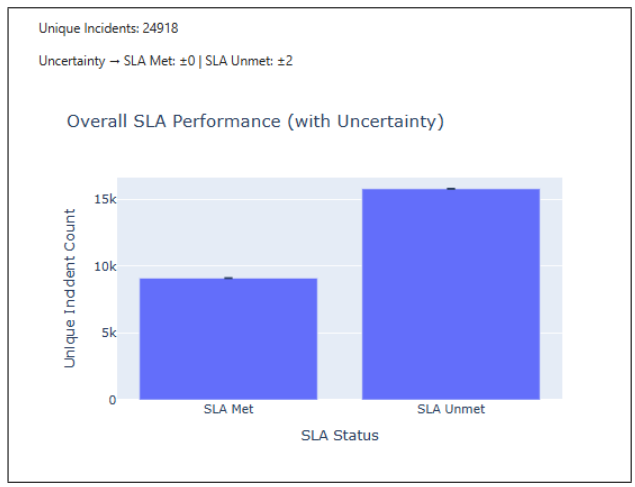


Fig. 1. Overall SLA compliance showing SLA met vs breached incidents

As shown in Fig. 1, the number of unique incidents that breached the SLA is approximately 1.7× higher than the number of incidents that met the SLA, indicating a significant imbalance in SLA compliance. This disparity suggests that certain incident characteristics may influence whether an SLA is met or breached. To better understand the underlying causes, we next examine how attributes such as $priority$, $urgency$, and $impact$ relate to $made_sla$, with the goal of identifying factors that contribute to SLA breaches and opportunities for improvement.

2) Effect of Priority, Impact, and Urgency on SLA Compliance: To understand how incident severity influences SLA outcomes, we analyze the relationship between $priority$, $impact$, $urgency$, and $made_sla$. For each level p of a given attribute (e.g., priority level), we compute the fraction of unique incidents that met and breached the SLA. Let N_p denote the total number of unique incidents at level p, $N_{met,p}$ denote the number of unique incidents that met the SLA, and $N_{unmet,p}$ denote

the number of incidents that breached the SLA. The corresponding fractions are defined as:

$$f_{met}(p) = \frac{N_{met,p}}{N_p}, \quad f_{unmet}(p) = \frac{N_{unmet,p}}{N_p}$$

Since every incident must either meet or breach the SLA, these fractions satisfy:

$$f_{met}(p) + f_{unmet}(p) = 1$$

To ensure transparency in progressive computation, we display uncertainty and stability as tooltips in all heatmaps. For a given attribute level, uncertainty is defined as

$$uncertainty_{p,sla} = |count_{current} - count_{previous}|$$

where $count_{current}$ and $count_{previous}$ represent the cumulative number of unique incidents with a given SLA outcome in the current and previous batches, respectively. Stability is then computed as

$$stability = 1 - \frac{uncertainty}{batch_size}.$$

These metrics quantify how much counts change between successive batches of 50 incidents, helping users assess the reliability of the visualized patterns. High stability indicates convergence, giving confidence that the trends reflect consistent data.

Fig. 2. Heatmap representing $made_sla$ vs priority.

As shown in Fig. 2, Fig. 3, and Fig. 4 incidents with higher severity—represented by lower numerical values of $priority$, $impact$, and $urgency$—exhibit a higher fraction of SLA compliance, indicating that critical incidents receive immediate attention. However, these high-severity incidents constitute a relatively small portion of the overall workload, while a larger share of SLA breaches originates from medium and low severity incidents. This observation

Fig. 3. Heatmap representing $made_sla$ vs impact.

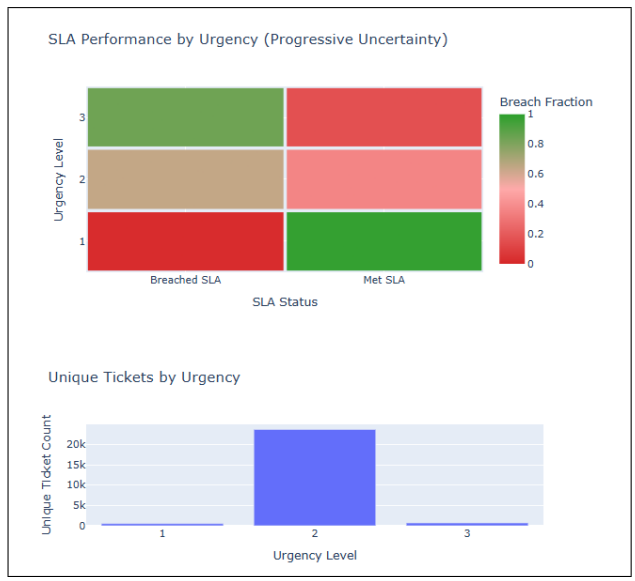


Fig. 4. Heatmap representing $made_sla$ vs urgency.

raises an important question: are these breaches primarily caused by workload distribution, or do they reflect performance limitations of the assigned support groups? To answer this, we next introduce a group capability score to evaluate how effectively each group resolves its assigned incidents.

3) Group Capability Score Analysis: To evaluate the performance of each assigned group and detect potential workload imbalances, we define a **severity-based capability score** for individual incidents:

$$S = \frac{10}{\log(priority + impact + urgency)}$$

where $priority$, $impact$, and $urgency$ are numerical values of the incident attributes. Higher-severity incidents (lower numerical values) receive higher weights.

For SLA outcomes, we define:

$$S_{met} = \frac{10}{\log(priority + impact + urgency)},$$
$$\text{if } made_sla = \text{false (SLA met)}$$

$$S_{breach} = \frac{10}{\log(priority + impact + urgency)},$$
$$\text{if } made_sla = \text{true (SLA breached)}$$

To control the penalty effect, we introduce a factor β, which is empirically chosen to limit the penalty to a fraction of the incident weight. In this work, we set $\beta = 0.15$ so that SLA breaches reduce the score moderately without outweighing the contribution from successfully resolved incidents.

The group capability is updated incrementally for each assigned incident:

$$\text{capability}_{group} = \sum_{i \in \text{SLA met}} S_{met}^{(i)} - \sum_{j \in \text{SLA breached}} \beta \cdot S_{breach}^{(j)}$$

Finally, we ensure each individual score remains non-negative:

$$\text{capability} = \max(0, \text{capability})$$

Summing the scores across all incidents per group provides a measure of **group performance**, highlighting workload distribution, performance bottlenecks, and potential improvements.

To quantify the **stability of capability over time**, we define the uncertainty for a group:

$$uncertainty_G = |G_{current\ batch} - G_{previous\ batch}|$$

where $G_{current\ batch}$ and $G_{previous\ batch}$ are the cumulative capability scores for the current and previous batches of incidents. Low uncertainty indicates that the group's performance has stabilized, capturing convergence trends and allowing confident operational decisions.

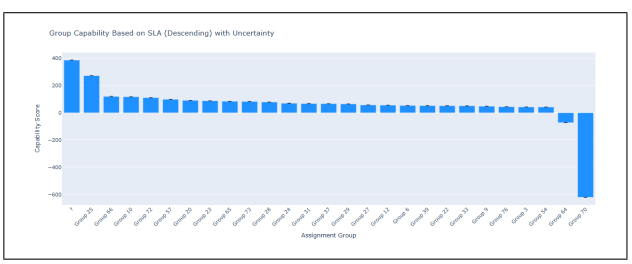


Fig. 5. Assignment Group vs Capability Score.

From Fig. 5, we observe that Group 70 initially performs very well, but as it is assigned a large number of incidents (9,400), its capability score begins to decrease, indicating that even high-performing groups can experience reduced efficiency under heavy workload. In contrast, Group 64 consistently underperforms, with only a small fraction of incidents meeting SLA targets, suggesting inherent performance limitations. Overall, most groups resolve approximately half of their incidents within the target SLA. These observations highlight the need to balance incident assignments and optimize group performance. One effective strategy is an epsilon-greedy assignment: with probability ϵ, assign incidents randomly to explore group capacity, and with probability $1-\epsilon$, assign incidents greedily to the best-performing group, thus improving SLA compliance while preventing overload on top-performing groups.

4) Top Categories by SLA Compliance: To understand how incident categories influence SLA outcomes, we analyze the distribution of unique incidents across categories using a bar chart. The x-axis represents number of unique incidents, and the y-axis shows the top 5 incident categories. This visualization identifies which categories contribute the most to overall workload and SLA performance.

The error associated with each category is computed using the standard error formula:

$$SE = \sqrt{y\left(1 - \frac{y}{N}\right)}$$

where y is the number of unique incidents observed in the category, and N is the total number of unique incidents observed across all categories.

This error represents the statistical uncertainty in the observed count due to finite sampling. Since each incident can either belong to the category or not, it can be modeled as a Bernoulli trial, where the probability of belonging to the category is given by $\frac{y}{N}$. The standard error therefore quantifies the uncertainty in estimating this probability from the observed data. Categories with fewer incidents have higher uncertainty, while categories with larger incident counts have lower uncertainty and more reliable estimates.

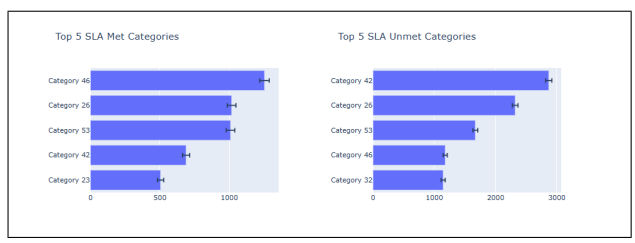


Fig. 6. Top 5 categories by SLA compliance.

As shown in Fig. 6, categories 42, 53, 26, and 46 account for the largest share of incidents, contributing significantly to both SLA compliance and SLA breaches. One practical approach to increase overall SLA performance is to assign dedicated or specialized support groups to these high-volume categories.

5) Priority Distribution Among Major Categories: The analysis of priority distribution shows that priority 3 incidents have the highest counts, leading to most SLA

breaches in these major categories (Fig. 7). A possible improvement is to train dedicated groups for these high volume categories.

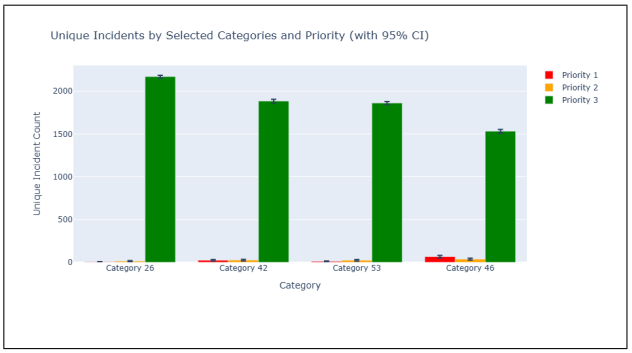


Fig. 7. Priority distribution in top 4 categories.

The uncertainty in the incident counts is represented using a 95% confidence interval derived from the binomial distribution. Let y denote the number of unique incidents in a given category (or priority), and let n denote the total number of unique incidents considered within that category. The observed proportion is defined as

$$\hat{p} = \frac{y}{n}$$

The standard error of this proportion is computed as

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

The 95% confidence interval is then obtained as

$$CI_{95\%} = y \pm 1.96 \cdot SE \cdot n$$

This interval represents the range within which the true underlying incident count is expected to lie with 95% confidence. In other words, if the incident sampling and observation process were repeated multiple times, approximately 95% of the computed confidence intervals would contain the true incident count. The confidence interval therefore quantifies the statistical uncertainty arising from finite observations. Narrow intervals indicate more reliable and stable estimates, while wider intervals reflect greater uncertainty due to fewer observations or higher variability.

6) Knowledge Base Usage Across Major Categories: The analysis of knowledge base usage for major categories is shown in Fig. 8. For the majority of unique incidents in these major categories, a knowledge base document was not used, which likely contributed to SLA targets not being met. Knowledge was not used possibly due to its non-existence or difficulty in locating it. After training dedicated groups, they can create a root-cause analysis (RCA) for each incident and its resolution, enabling future use of the knowledge base to improve SLA compliance.

The uncertainty associated with the unique incident counts for knowledge base usage is quantified using a 95% confidence interval based on the binomial approximation.

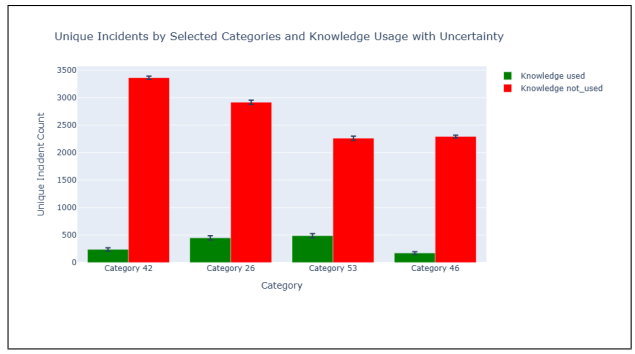


Fig. 8. Knowledge base utilization by category.

Let y denote the number of unique incidents in a category where the knowledge base was either used or not used, and let n denote the total number of unique incidents in that category. The observed proportion is defined as

$$\hat{p} = \frac{y}{n}$$

The standard error of this proportion is given by

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

To express uncertainty in terms of incident counts rather than proportions, the confidence interval is scaled to the count domain as

$$CI = z \cdot SE \cdot n$$

where $z = 1.96$ corresponds to the critical value for a 95% confidence level under the normal approximation. The incident count is therefore represented as

$$y \pm CI$$

This 95% confidence interval represents the range within which the true underlying incident count associated with knowledge base usage is expected to lie with 95% confidence. In practical terms, if the observation process were repeated multiple times, approximately 95% of the computed intervals would contain the true incident count. Narrower intervals indicate more reliable estimates, while wider intervals reflect greater statistical uncertainty due to smaller sample sizes or higher variability.

7) Impact of Knowledge Base Usage on SLA Compliance: To quantify the effect of knowledge base (KB) usage on meeting SLA targets, we organize incidents into a 2×2 contingency table:

	SLA Met	SLA Unmet
KB Used	a	b
KB Not Used	c	d

Here, a = number of incidents where KB was used and SLA was met, b = KB used and SLA breached, c = KB not used and SLA met, d = KB not used and SLA breached.

The **Odds Ratio (OR)** measures the odds of meeting SLA when KB is used relative to when it is not:

$$OR = \frac{a \cdot d}{b \cdot c}$$

- $OR = 1$: KB usage has no effect on meeting SLA - $OR > 1$: Using KB increases the odds of meeting SLA - $OR < 1$: Using KB decreases the odds of meeting SLA

The **Relative Probability / Risk Ratio (RP)** measures the probability of meeting SLA when KB is used relative to not using KB:

$$RP = \frac{a/(a+b)}{c/(c+d)}$$

- $RP = 1$: KB usage does not change the probability of meeting SLA - $RP > 1$: KB usage increases the probability of meeting SLA - $RP < 1$: KB usage decreases the probability of meeting SLA

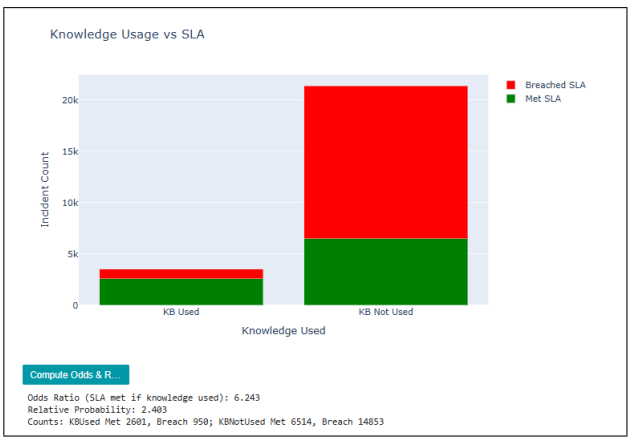


Fig. 9. Knowledge Used Vs SLA.

As seen from Fig. 9, using KB increases the odds of meeting SLA by 6.243 times and increases the probability of meeting SLA by 2.403 times. This reinforces the recommendation to train dedicated groups on major categories 42, 53, 26, and 46, and to document incident resolutions in the knowledge base for future reference, thereby improving SLA compliance.

B. Incident concentration trend analysis

In this section, we aim to identify specific days of the week and time intervals during the day when incident volumes peak. These insights allow for strategic allocation of resolution groups, improving incident response efficiency and enhancing overall operational capacity. Additionally, we investigate whether higher incident volumes correlate with an increased likelihood of SLA breaches.

1) Incident distribution by day of week: The distribution of unique incidents by weekday is illustrated in Fig. 10. On the x-axis, the days of the week are represented (Monday through Sunday), while the y-axis shows the number of unique incidents occurring on each day, aggregated over several weeks.

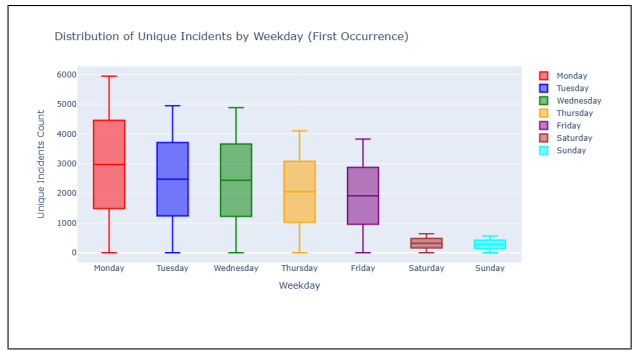


Fig. 10. Distribution of unique incident by weekdays.

As seen from Fig. 10, incident volume is highest on Monday and gradually decreases as the week progresses, indicating a surge in operational activity at the start of the week. The lowest number of incidents occurs on Saturday and Sunday, likely reflecting reduced business operations and lower system usage during weekends. A plausible explanation for this pattern is that operational and business activities typically peak at the beginning of the week as employees resume work, pending tasks are executed, and system interactions increase. As the week progresses, operational intensity stabilizes or decreases, resulting in fewer new incidents, while reduced weekend activity naturally produces lower incident counts.

Next, we examine the time-of-day patterns to identify when incidents peak during the day.

2) Running average of incident distribution by hour of day: To identify the time intervals when incident volumes peak, we compute the hourly distribution of incidents. For each day, we count the number of incidents occurring in each hour (0th to 23rd hour). To smooth fluctuations across multiple days, we maintain a progressive running average and standard deviation for each hour.

Let $I_{h,d}$ denote the number of incidents observed in hour h on day d, and let $\overline{I}_h^{(d-1)}$ be the running average for hour h after $d-1$ days. The running average is updated incrementally as follows:

$$\overline{I}_h^{(d)} = \frac{(d-1) \cdot \overline{I}_h^{(d-1)} + I_{h,d}}{d}$$

The running standard deviation $\sigma_h^{(d)}$ is updated incrementally as:

$$\sigma_h^{(d)} = \sqrt{\frac{(d-2) \cdot (\sigma_h^{(d-1)})^2 + (I_{h,d} - \overline{I}_h^{(d-1)}) \cdot (I_{h,d} - \overline{I}_h^{(d)})}{d-1}}$$

Using the running average and standard deviation, we compute the 95% confidence interval (CI) for the mean incident count:

$$CI_h^{(d)} = \overline{I}_h^{(d)} \pm 1.96 \cdot \frac{\sigma_h^{(d)}}{\sqrt{d}}$$

Here:

- $I_{h,d}$ = incident count in hour h on day d
- $\overline{I}_h^{(d)}$ = running average incident count for hour h after d days
- $\sigma_h^{(d)}$ = running standard deviation of incident counts for hour h
- d = number of days processed

The shaded band around the running average curve represents the 95% confidence interval, conveying the uncertainty of the mean incident count at each hour. A narrower band indicates stable hourly patterns, while a wider band shows higher variability and less certainty about the expected incident volume.

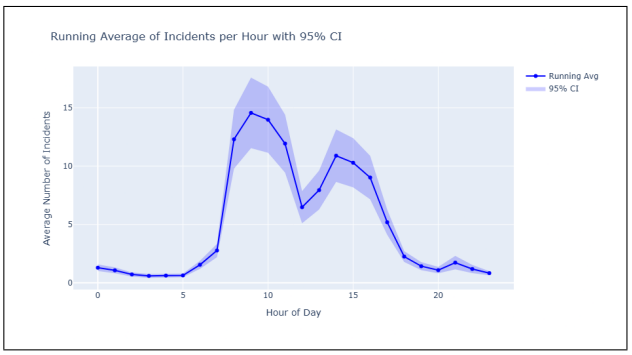


Fig. 11. Running average of unique incident count by hour of day.

From Fig. 11, we observe that incident volumes tend to peak between 08:00–11:00 and 13:00–16:00, indicating that these time intervals correspond to the highest operational activity during the day. This pattern suggests that strategically allocating resolution teams during these peak hours can improve response capacity and support better SLA adherence. The concentration of incidents during standard business hours highlights a clear relationship between user activity and incident occurrence, while the dip around midday may reflect temporary declines in system usage, possibly corresponding to organizational break periods. Next, we examine whether this hourly pattern is consistent across all weekdays.

3) Hourly Incident Distribution by Weekday: From Fig. 12, we examine the unique incident counts per hour for each weekday to assess whether the pattern observed in the running average holds consistently. The analysis reveals that, for weekdays, the hourly trend is largely consistent, with incidents peaking between 08:00–11:00 and 13:00–16:00. Saturdays and Sundays are exceptions due to reduced organizational activity. Overall, this observation reinforces and solidifies the trend identified by the running average, confirming the recurring peak periods during standard business hours.

4) Category Distribution During Peak Incident Hours: To support efficient workforce allocation during peak operational intervals, we analyze the distribution of incident categories during the identified peak hours of 08:00–11:00 and 13:00–16:00. This helps determine which categories contribute most to incident volume during high-load periods.

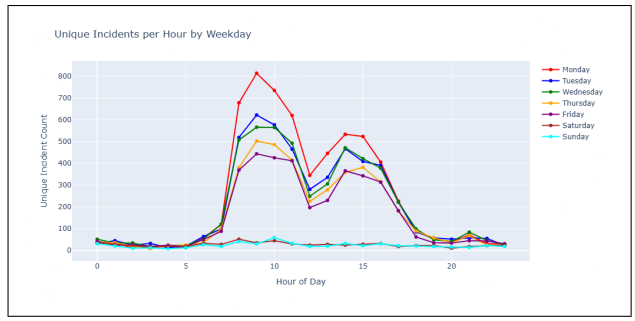


Fig. 12. Unique incident count by hour of the day distributed among weekdays.

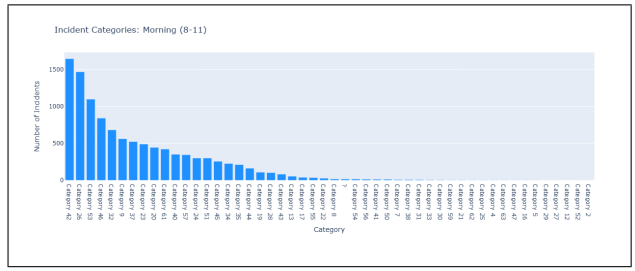


Fig. 13. Incident categories: Morning (8.00-11.00).

From Fig. 13, and Fig. 14, we observe that categories 42, 23, 53, and 46 consistently appear most frequently during both peak intervals. This reinforces our earlier findings that these categories dominate overall incident volume and are major contributors to operational workload.

This insight suggests that assigning dedicated resolution groups and allocating additional resources to these high-frequency categories during peak hours can improve SLA compliance.

5) SLA Breaches by Time of Day: We now analyze the temporal distribution of SLA breaches to identify when breaches occur most frequently. From Fig. 15, we observe that the number of SLA breaches is proportional to the incident arrival volume within each time slot. Specifically, the 08:00–11:00 interval exhibits the highest number of SLA breaches, which corresponds to the peak incident inflow during this period, whereas the 13:00–16:00 interval shows comparatively fewer breaches due to relatively lower incident volume.

This insight suggests that allocating additional resolution resources during peak morning hours can help reduce

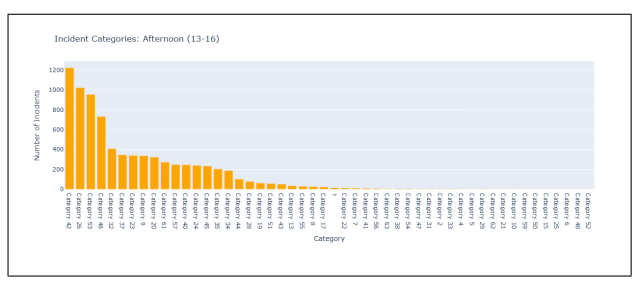


Fig. 14. Incident categories: Afternoon (13.00-16.00).

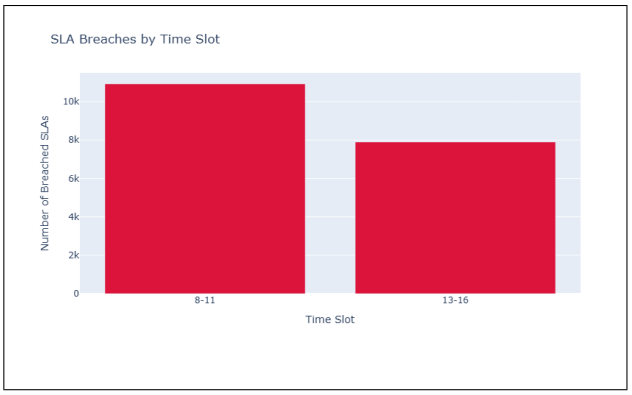


Fig. 15. SLA breaches by timeslot.

SLA breaches and improve overall incident resolution performance.

IV. Forensic Analysis

A. Escalation Impact on Resolution Time

Figure 16 presents the escalation dashboard, highlighting the relationship between reassignment depth and incident resolution time. The results reveal a clear monotonic trend: resolution time increases consistently with higher values of `reassignment_count`. Incidents reassigned three or more times exhibit substantially higher median resolution durations compared to those resolved with fewer handoffs.

This finding indicates that repeated escalations introduce cumulative coordination overhead and inter-team handoff delays, significantly slowing the resolution process. Escalation depth therefore emerges as a critical structural driver of operational inefficiency and prolonged incident handling.

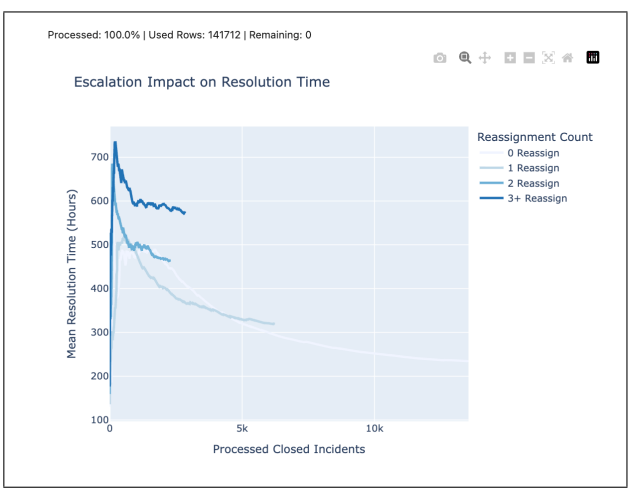


Fig. 16. Escalation impact on resolution time.

B. Priority-Level Performance Assessment

Figure 17 illustrates the relationship between incident priority and resolution time. Under SLA hierarchy assumptions, Priority 1 incidents are expected to be resolved faster than lower-priority cases. However, the visualization reveals that Priority 1 incidents do not consistently achieve shorter resolution times compared to Priority 2 incidents.

This unexpected inversion suggests potential workload saturation, resource bottlenecks, or increased operational complexity associated with high-priority cases. The finding motivates deeper forensic investigation to identify structural inefficiencies affecting priority-based resolution performance.

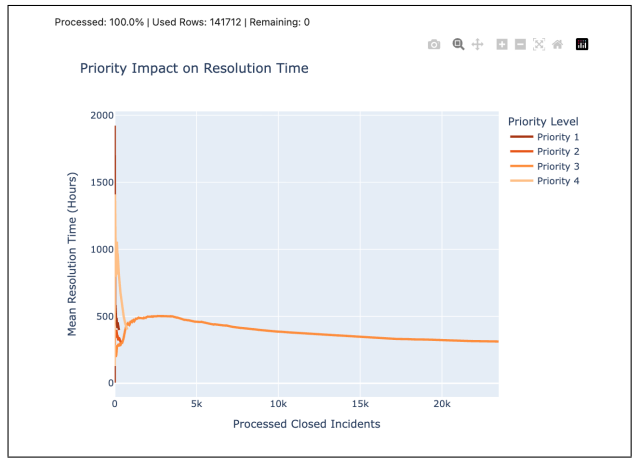


Fig. 17. Priority impact on resolution time.

C. SLA Compliance Analysis

Figure 18 illustrates SLA breach rates stratified by escalation depth. The visualization clearly demonstrates that SLA breach probability increases substantially as `reassignment_count` grows. Incidents undergoing multiple escalations exhibit significantly higher breach rates compared to those resolved within a single assignment cycle.

This structural trend confirms that escalation depth not only prolongs resolution time but also directly undermines SLA compliance performance.

Figure 19 presents SLA compliance distribution across priority levels. Although higher-priority incidents are expected to receive expedited handling, breach rates remain elevated across all priority categories. This indicates that SLA violations are not merely a consequence of urgency misclassification, but rather reflect broader systemic inefficiencies within the incident management workflow.

Figure 20 further analyzes SLA compliance across major incident categories. Certain high-volume categories exhibit disproportionately high breach rates, suggesting localized process bottlenecks. These results indicate that category-specific workflow optimization or specialized resolution groups could substantially improve overall SLA performance.

D. Dimensionality Reduction and Structural Clustering

To identify latent structural patterns in incident behavior, we applied Principal Component Analysis (PCA) to the standardized numeric feature

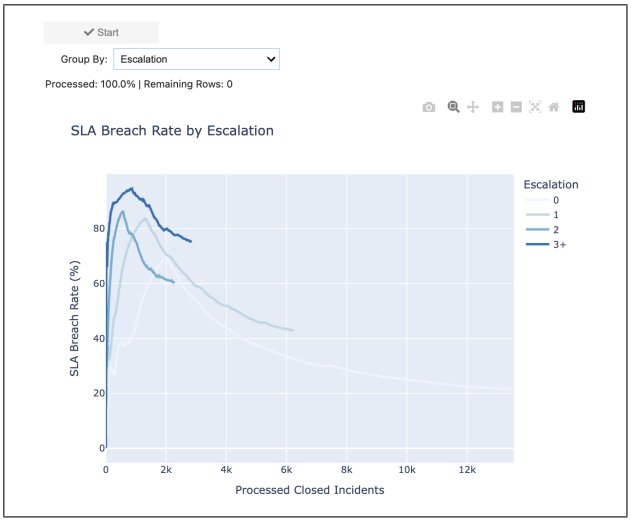


Fig. 18. SLA breach rate by escalation depth.

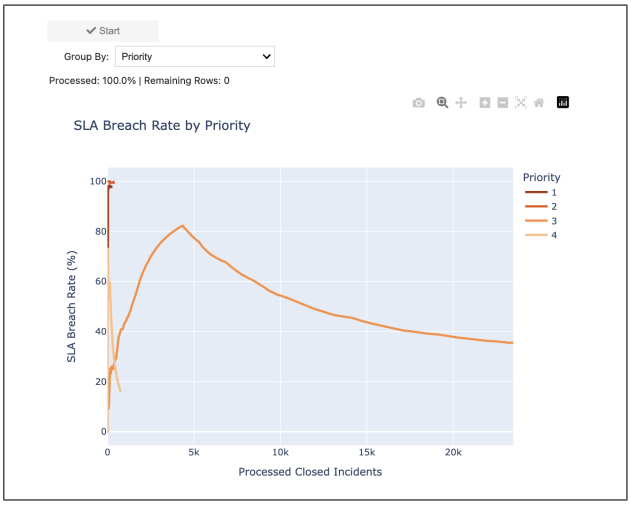


Fig. 19. SLA breach rate by priority level.

Fig. 20. SLA breach rate across major incident categories.

vector: `[reassignment_count, reopen_count, sys_mod_count, priority, impact, urgency]`.

The objective was to project high-dimensional operational attributes into a two-dimensional embedding while preserving dominant variance directions.

Figure 21 shows the PCA projection of incidents in the space defined by the first two principal components. Rather than random dispersion, the visualization reveals distinguishable structural regions. Routine incidents cluster tightly within a low-complexity region characterized by minimal reassignment and modification activity. Moderately complex incidents occupy transitional regions, while high-complexity outliers appear more dispersed and are often associated with elevated system modification counts and repeated escalations.

Fig. 21. PCA projection of incidents highlighting structural clusters.

The presence of separable clusters confirms that incident resolution behavior follows correlated operational manifolds rather than independent attribute variation. Dimensionality reduction therefore provides structural evidence that escalation depth, system modifications, and urgency attributes interact collectively in shaping resolution outcomes.

E. May 2016 Forensic Case Study

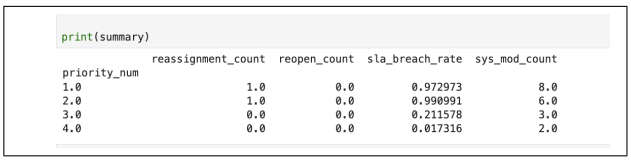
A pronounced anomaly was detected during May 2016 when analyzing resolution time distributions across priority levels. Figure 22 presents a boxplot comparison between Priority 1 and Priority 2 incidents during this period. Contrary to SLA expectations, Priority 1 incidents exhibit a higher median resolution time (190.17 hours) than Priority 2 incidents (141.43 hours).

This inversion contradicts urgency-based service assumptions, where higher-priority incidents should receive faster resolution. Additionally, Priority 1 incidents display greater dispersion, indicating increased variability in handling performance during this interval.

To determine whether this behavior was isolated or structural, a global comparison was performed across the full dataset. Figure 23 summarizes overall resolution time distributions by priority level. Unlike the May-specific inversion, the global pattern follows a more consistent

Fig. 22. Resolution time comparison between Priority 1 and Priority 2 incidents (May 2016).

ordering, where higher-priority incidents tend to exhibit slightly lower median resolution times. Nevertheless, substantial variance persists across all priority categories.



```
print(summary)
                reassignment_count  reopen_count  sla_breach_rate  sys_mod_count
priority_num
1.0                            1.0           0.0         0.972973            8.0
2.0                            1.0           0.0         0.990991            6.0
3.0                            0.0           0.0         0.211578            3.0
4.0                            0.0           0.0         0.017316            2.0
```

Fig. 23. Overall resolution time distribution across priority levels.

Having established that the inversion was temporally localized, further investigation examined potential complexity drivers. Escalation depth and reopen cycles were analyzed; however, no abnormal increase in `reassignment_count` or `reopen_count` was observed for Priority 1 incidents during May 2016.

Subsequent analysis revealed elevated `sys_mod_count` values for high-priority incidents in this period. The system modification count reflects the number of workflow or state changes applied to an incident record, and higher values typically indicate increased coordination effort and internal process adjustments.

These findings suggest that the anomaly was not primarily caused by escalation mismanagement, but rather by increased handling complexity and organizational strain. The concurrent observation of widespread SLA breaches during this interval further supports the hypothesis of temporary systemic overload.

The May 2016 case study demonstrates the importance of integrated visual analytics. The anomaly would not have been detectable through isolated metric inspection; instead, it emerges from coordinated analysis of resolution distributions, priority ordering, escalation metrics, and structural complexity indicators. This illustrates how forensic visual analytics can uncover temporally localized inefficiencies that remain obscured within aggregate performance statistics.

V. CONCLUSION

This work demonstrates the effectiveness of progressive visual analytics in enabling interactive and incremental exploration of incident data, allowing meaningful insights to be obtained without waiting for complete computation. By leveraging the ProgressiVis framework, we were able to continuously monitor evolving patterns, visualize trends, and support real-time analytical reasoning. This approach improves responsiveness, enables early detection of operational patterns, and supports informed decision-making. Overall, progressive visual analytics provides a powerful framework for scalable, efficient, and adaptive analysis of large datasets.

REFERENCES

[1] J. Lim and C. Song, "Clustering IT Incidents Using K-Means: Improving Incident Response Time in Service Management," *Sinkron*, 2024. [Online]. Available: `https://jurnal.polgan.ac.id/index.php/sinkron/article/view/14822/3287`.

[2] R. Anggraeni, F. Alzami, A. Nurhindarto, S. Budi, R. A. Megantara, I. Rizqa, and M. Muslih, "Graph-Based Analytical Approach to Identifying Substitute Human Resources: Integrating Individual Capabilities and Group Dynamics," *Systems*, MDPI, 2024. [Online]. Available: `https://www.mdpi.com/2079-8954/14/1/32`.

[3] UCI Machine Learning Repository. (2019). Incident Management Process Enriched Event Log Dataset. [online]. Available: `https://archive.ics.uci.edu/dataset/498/incident+management+process+enriched+event+log`. [Accessed 26 Feb. 2026].

[4] ProgressiVis Documentation. (2024). ProgressiVis: A Toolkit for Progressive Visual Analytics. [online]. Available: `https://progressivis.readthedocs.io/en/latest/`. [Accessed 26 Feb. 2026].

[5] ProgressiVis GitHub Repository. (2024). ProgressiVis Progressive Visual Analytics Tool. [online]. Available: `https://github.com/progressivis/progressivis`. [Accessed 26 Feb. 2026].

[6] Fekete, J.-D., et al. (2018). Progressive Visual Analytics. Eurographics Digital Library. [online]. Available: `https://diglib.eg.org/items/c146a716-7b3f-469a-95a2-a354d349389e`.

[7] Eurographics Association. (2018). Progressive Visual Analytics: Challenges and Opportunities. [online]. Available: `https://diglib.eg.org/items/c146a716-7b3f-469a-95a2-a354d349389e`. [Accessed 26 Feb. 2026].